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To Sense or Not To Sense: An AoI Perspective in Energy-Harvesting Networks

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***Abstract*—This paper investigates the age of information (AoI) in energy harvesting (EH) wireless networks employing Carrier Sense Multiple Access (CSMA). We model the stochastic energy harvesting process and the channel state evolution, and derive closed-form expressions for the average AoI. The optimal packet generation rate and corresponding minimum average AoI are determined under the assumption of an infinite energy buffer. Moreover, we compare the optimal average AoI performance of EH-CSMA and EH-Aloha systems. The results show that CSMA significantly reduces the average AoI, particularly in dense networks, while Aloha may outperform CSMA in scenarios with high propagation delays.**

***Index Terms*—Age of information, Energy harvesting, CSMA, Aloha**

I. Introduction

Ensuring the timely delivery of status updates is essential for emerging Internet of Things (IoT) applications such as smart wearable devices, environmental monitoring, and smart sensing. Traditional metrics like throughput and delay are insufficient to characterize the freshness of information. In response, the age of information (AoI) was introduced to measure the time elapsed since the latest successfully received update was generated [1]. AoI has emerged as a critical metric for evaluating information freshness in systems requiring timely data delivery [2].

Compared to centralized access schemes that depend on global coordination and scheduling, random access provides a more scalable and flexible solution, which is more suitable for intelligent IoT networks. Among these, Carrier Sense Multiple Access (CSMA) is widely adopted for its ability to mitigate collisions by sensing the channel before transmission, while maintaining a distributed and contention-based access behavior. Several studies have investigated the AoI performance in CSMA networks [3]–[5]. In [3], authors derive closed-form expressions for the average AoI in unsaturated CSMA networks. [4] derives average AoI expressions for deterministic arrivals and an upper bound for stochastic arrivals. The worst-case AoI is examined in [5], highlighting the importance of tuning the contention window size. However, applying these results in practice presents limitations since the aforementioned studies do not account for the stringent energy constraints of devices.

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In practical intelligent IoT networks where devices are typically energy-constrained, the transmission behavior is inherently affected by energy availability, which makes the existing analytical frameworks for AoI under infinite power supply assumptions insufficient when applied to realistic settings. Our previous work [6] and [7] attempt to evaluate the performance of energy-constrained IoT networks. Particularly, [6] analyzes the trade-off between AoI and energy consumption in Poisson networks, revealing how sampling and transmission strategies affect this balance. [7] examines connection-based and packet-based Aloha under battery limits, aiming to maximize energy efficiency over the device lifetime by balancing energy cost and throughput. Our results suggest that battery-powered systems require a relatively high energy budget to maintain desirable performance.

To address power limitations more sustainably, energy harvesting (EH) has emerged as a promising solution for powering IoT nodes by collecting energy from the environment. In EH-enabled networks, the transmission process is tightly coupled with the stochastic energy arrival process, thus requiring analytical frameworks beyond those used in conventional systems. Recent research efforts have explored the AoI in EH-Aloha systems [8]–[10]. For instance, [8] demonstrates through theoretical analysis and simulations how different transmission strategies impact AoI in EH-Aloha network. [9] derives the average AoI expressions for EH-Aloha networks with general energy buffers and optimizes the update rate accordingly. A hybrid approach in [10] combines batch-service Markov chains with stochastic geometry to jointly optimize the update rate and packet length for minimizing average AoI. Most of these studies focus on Aloha protocols, while investigations into EH-CSMA networks remain limited. For example, [11] analyzes EH-CSMA networks employing IEEE 802.11 protocols and evaluates a modified distributed coordination function using a three-dimensional Markov chain model. However, the analysis is limited to throughput and delay performance. To the best of our knowledge, the AoI performance in EH-CSMA networks remains an open and important research direction.

This gap is particularly important given the fundamental differences between Aloha and CSMA in their channel access mechanisms and their applicability to different IoT scenarios. Existing studies have established a performance comparison framework for both protocols without energy constraints. In particular, [12] assesses their throughput, delay, and stability using the HOL-packet approach. And [13] emphasizes that

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Aloha is more suitable for M2M communication, particularly in scenarios involving short packets and a large number of low-power devices. Furthermore, [14] aims to minimize the peak AoI in CSMA and compares it with Aloha. Given the constraints of energy harvesting, a natural but insufficiently explored question is to systematically compare the AoI performance of EH-Aloha and EH-CSMA across various network configurations, in order to determine their respective suitability for different types of applications.

To address these questions, this paper characterizes and optimizes the average AoI in EH-CSMA networks and compares the optimal average AoI with that of EH-Aloha networks, thereby revealing the criteria under which a sensing mechanism becomes beneficial. Our contributions are summarized as follows:

- We derive closed-form expressions for the average AoI, and further obtain the optimal average AoI and the corresponding optimal packet generation rate under the assumption of an infinite energy buffer.
- We conduct simulations to validate our analytical results, demonstrating a strong agreement between theoretical expressions and simulation results across various network configurations.
- We compare the optimal average AoI in EH-CSMA and EH-Aloha networks and find that CSMA significantly outperforms Aloha in energy-constrained and low-delay networks, whereas Aloha may perform better under delay-tolerant scenarios.

II. System Model

A. Network Configuration and Transmission Protocol

We consider a CSMA network consisting of n EH nodes, each equipped with an identical EH sensor. All nodes are capable of sensing the channel and attempting to access a common receiver.

The time is slotted and divided into mini-slots of length a, where a represents the ratio of the propagation delay required for channel sensing to the packet transmission time. Both successful transmissions and collision detections are assumed to last one time slot. This implies that the nodes operate in the half-duplex mode and are unaware of the collision during the transmission of the whole packet. All nodes are assumed to be synchronized. Within one time slot, a node independently harvests one unit of energy with constant probability δ, following a Bernoulli process.[1] Each node can store up to B units of energy. Once the energy buffer is full, any subsequent harvested energy is discarded.

Moreover, we adopt the generate-at-will strategy, in which a fresh packet is generated immediately before transmission. Each attempted transmission consumes one unit of energy, while the energy consumed for channel sensing is negligible compared to that required for transmission.[2] When a node detects that the channel is idle and its energy buffer has at least one unit of energy, it generates a fresh packet with probability q and attempts to transmit at the beginning of a mini-slot; otherwise, it remains inactive.

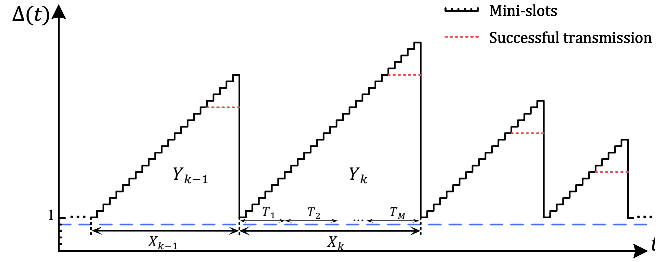


Fig. 1. Example of the AoI trace in a CSMA network with the mini-slot length $a = 0.2$.

Note that all nodes share the same spectrum for transmissions and may interfere with one another. Therefore, we utilize the collision model to characterize the co-channel interference, which means that a transmission is successful only if no concurrent transmission occurs over the channel.

B. Performance Metric

We leverage the age of information (AoI) as a performance metric to measure information freshness from the receiver's perspective in a CSMA network. AoI is defined as the time elapsed since the generation of the latest successfully received packet. Fig. 1 illustrates the evolution of AoI over time in a CSMA network, where the mini-slot length is $a = 0.2$. The AoI increases by a per mini-slot and resets to one after a successful transmission. The red dashed line indicates successful transmissions, with each transmission lasting 5 mini-slots, as determined by $\frac{1}{a}$. The evolution of AoI can be expressed as

$$\Delta(t) = t - G(t). \tag{1}$$

where $G(t)$ is the generation time of the latest packet received at time t.

III. Analysis

This paper focuses on the performance analysis of the average AoI in EH-CSMA networks. Based on the above analysis, the average AoI can be expressed as

$$\begin{aligned}\bar{\Delta} &= \lim_{t\to\infty} \frac{(1-a)t + \sum_{k=1}^{K} Y_k}{t} = \lim_{t\to\infty} \frac{K}{t}\frac{\sum_{k=1}^{K} Y_k}{K} + 1 - a \\ &= \frac{\mathbb{E}\left[\frac{1}{2}X^2 + \frac{1}{2}X\right]}{\mathbb{E}[X]} + 1 - a = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]} + \frac{3}{2} - a,\end{aligned} \tag{2}$$

[1] In practical EH systems, energy arrival rates are typically low and sporadic, depending on the ambient environment and the specific harvesting technique employed. For instance, RF sources typically provide 10–100 μW at moderate ranges [15], while indoor and outdoor solar sources yield approximately 100 μW to 100 mW under typical conditions [16]. Suppose a packet transmission consumes approximately 60 μJ over 1 ms, and energy arrives 6 μJ/ms; then, the normalized energy arrival rate δ can be set to 0.1.

[2] The energy consumed during channel sensing is typically two to three orders of magnitude lower than that required for data transmission. For example, clear channel assessment in IEEE 802.15.4 lasts for 128–300 μs, consuming 0.06–0.13 μJ. In contrast, transmitting a 20–100 byte packet costs 60–190 μJ over 1.2–3.7 ms, depending on the transmit power and packet size [17] [18]. Thus, the energy cost of sensing can be neglected.

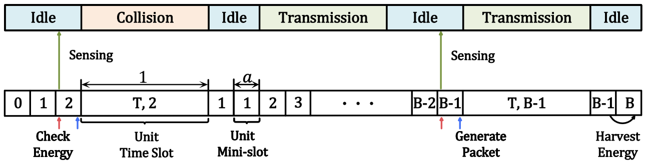


Fig. 2. States of channel and node in CSMA network. $x = \frac{1}{a}$.

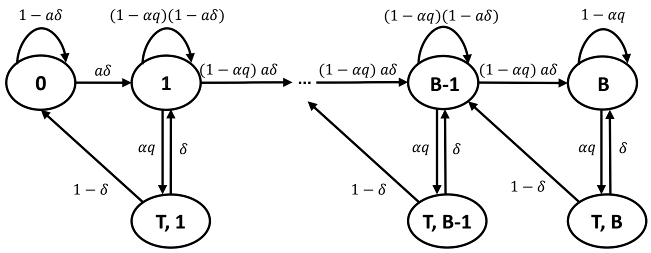


Fig. 3. An embedded Markov chain characterizing the dynamics of energy harvesting and consumption.

where the time between the $(k-1)$-th and the k-th successful transmissions X_k is the sum of the durations of M attempted transmissions, i.e., $X_k = \sum_{i=1}^{M} T_i$. Note that M is a random variable that follows a geometric distribution with parameter p. Therefore, the first and second moments of X_k can be expressed as follows

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{M} T_i \left(1-p\right)^{M-1} p\right] = \frac{\mathbb{E}[T]}{p}, \tag{3}$$

and

$$\begin{aligned}\mathbb{E}[X^2] =& \mathbb{E}\left[\left(\sum_{i=1}^{M} T_i\right)^2 \left(1-p\right)^{M-1} p\right] \\ =& \frac{\mathbb{E}[T^2]}{p} + \frac{2\left(1-p\right)}{p^2}\left(\mathbb{E}[T]\right)^2,\end{aligned} \tag{4}$$

where T represents the time between two consecutive attempted transmissions and p denotes the steady-state probability of successful transmission. By substituting (3) and (4) into (2), the average AoI can be written as

$$\bar{\Delta} = \frac{\mathbb{E}\left[T^2\right]}{2\mathbb{E}[T]} + \frac{1-p}{p}\mathbb{E}[T] + \frac{3}{2} - a, \tag{5}$$

Therefore, we will derive expressions for p and the first and second moments of T in the subsequent analysis to obtain the average AoI.

A. Embedded Markov Chain Modeling

1) State Transition Characterization: CSMA network has three possible channel states: idle, successful transmission, and collision detection. Fig. 2 illustrates the channel and node states. When a node is in state $\{T, i\}$, it indicates that the node has stored i units of energy before transmission and is undergoing either a successful transmission or collision detection. When a node is in state i, it has stored i units of energy but is not transmitting. The red arrow represents the energy checking process, the green arrow represents the node sensing that the channel is idle, and the blue arrow represents packet generation. Once a packet is generated, the node starts transmission in the next mini-slot, transitioning from state i to state $\{T, i\}$, which lasts for $\frac{1}{a}$ mini-slots.

The state transition of the embedded Markov chain is shown in Fig. 3. As shown, the probability of harvesting one unit of energy in each mini-slot is approximately $a\delta$. This approximation aligns with the intuition behind the Poisson process, which suggests that in sufficiently small mini-slots or a small energy arrival rate, the probability becomes proportional to the length of the mini-slot.

2) Steady-State Probability: The steady-state probability distribution of the embedded Markov chain can be obtained as

$$\begin{cases} \pi_0 = (1-\alpha q)\phi^B \pi_B, & \\ \pi_i = \phi^{B-i}\pi_B, & 1 \le i \le B-1, \\ \pi_{T,i} = \alpha q \phi^{B-i}\pi_B, & 1 \le i \le B-1. \end{cases} \tag{6}$$

and

$$\pi_B = \frac{1-\phi}{(1+\alpha q)\left(1-\phi^B\right) + (1-\alpha q)\phi^B(1-\phi)}, \tag{7}$$

where $\phi = \frac{\alpha q(1-\delta)}{(1-\alpha q)a\delta}$. According to [12], the steady-state probability of sensing the channel idle α can be similarly derived as

$$\alpha = \frac{a}{1+a-p}. \tag{8}$$

The transition from state i to any other state (including itself) takes 1 mini-slot, while the transition from state $\{T, i\}$ to any other state takes $\frac{1}{a}$ mini-slots. Therefore, the real holding time (in units of time slots) at state i and $\{T, i\}$, is given by

$$\tau_i = a \;(0 \le i \le B), \text{ and } \tau_{T,i} = 1 \;(1 \le i \le B). \tag{9}$$

Consequently, the limiting state probabilities of the Markov renewal process are given by

$$\tilde{\pi}_i = \frac{\pi_i \cdot \tau_i}{\sum_{j\in\mathbb{S}} \pi_j \cdot \tau_j}, i \in \mathbb{S}, \tag{10}$$

where $\mathbb{S} = \{\{T,1\}, ..., \{T,B\}, 0, 1, ..., B\}$.

B. Transmission Success Probability

According to the above analysis, the transmission success probability p, given that the channel is sensed idle, can be expressed as

$$\begin{aligned} p =& (\Pr\{\text{node has energy without packet}|\text{channel is idle}\} \\ &+ \Pr\{\text{node has no energy}|\text{channel is idle}\})^{n-1} \\ =& \left(\frac{(1-q)\sum_{i=1}^{B}\tilde{\pi}_i + \tilde{\pi}_0}{\sum_{i=0}^{B}\tilde{\pi}_i}\right)^{n-1}. \end{aligned} \tag{11}$$

By substituting $\tilde{\pi}_i$ into (11), thus with a large number of nodes n, the steady-state transmission success probability for a large number of nodes n can be expressed as

$$p \approx \exp\left(-\frac{nq\left(1-\phi^B\right)}{(1-\alpha q)\phi^B(1-\phi) + 1 - \phi^B}\right). \tag{12}$$

C. Analysis of Attempt Transmission Interval

Then we focus on the time interval T between two consecutive transmission attempts, which is influenced by the energy remaining after the previous attempt. If no energy remains, the node must first harvest enough energy before attempting access again. Conversely, if sufficient energy is stored at the end of the previous attempt, the node only needs to wait for access before initiating a new transmission.

We decompose T into two independent components: $T = T_E + T_A$, where T_E denotes the time required to harvest sufficient energy, and T_A represents the waiting time for channel access once enough energy is available. Both T_E and T_A can be approximated by geometric distributions. Therefore, we have

$$\mathbb{E}[T_E] = \frac{1}{a\delta}\cdot a = \frac{1}{\delta}, \text{ and } \mathbb{E}[T_A] = \frac{a}{\alpha q}. \tag{13}$$

and

$$\mathbb{E}[{T_E}^2] = \frac{2-a\delta}{(a\delta)^2}a^2 = \frac{2-a\delta}{\delta^2}, \text{ and } \mathbb{E}[{T_A}^2] = \frac{2-\alpha q}{(\alpha q)^2}a^2. \tag{14}$$

Furthermore, we can derive the first and second moments of T as follows

$$\begin{aligned}\mathbb{E}[T] =& \frac{\tilde{\pi}_1(1-\delta)}{\sum_{i=1}^{B}\tilde{\pi}_i}\mathbb{E}[T_E+T_A] + \left(1-\frac{\tilde{\pi}_1(1-\delta)}{\sum_{i=1}^{B}\tilde{\pi}_i}\right)\mathbb{E}[T_A]\\ =& \frac{a}{\alpha q} + \frac{\phi^{B-1}\left(1-\phi\right)\left(1-\delta\right)}{1-\phi^B}\frac{1}{\delta},\end{aligned} \tag{15}$$

and

$$\begin{aligned}\mathbb{E}\left[T^2\right] =& \frac{\tilde{\pi}_1(1-\delta)}{\sum_{i=1}^{B}\tilde{\pi}_i}\mathbb{E}[(T_E+T_A)^2]\\ &+ \left(1-\frac{\tilde{\pi}_1(1-\delta)}{\sum_{i=1}^{B}\tilde{\pi}_i}\right)\mathbb{E}[{T_A}^2]\\ =& \frac{(2-\alpha q)a^2}{(\alpha q)^2} + \frac{\phi^{B-1}\left(1-\phi\right)\left(1-\delta\right)}{1-\phi^B}\\ &\cdot\left(\frac{2-a\delta}{\delta^2} + \frac{2a}{\alpha q\delta}\right).\end{aligned} \tag{16}$$

D. Average AoI Analysis

Now, by substituting (12), (15), and (16) into (5), the closed-form expression for the average AoI $\bar{\Delta}$ in EH-CSMA networks can be expressed as

$$\begin{aligned}\bar{\Delta} =& \frac{\frac{(2-\alpha q)a^2}{(\alpha q)^2} + \frac{\phi^{B-1}(1-\phi)(1-\delta)}{1-\phi^B}\left(\frac{(2-a\delta)}{\delta^2}+\frac{2a}{\alpha q\delta}\right)}{2\left(\frac{a}{\alpha q}+\frac{\phi^{B-1}(1-\phi)(1-\delta)}{\delta(1-\phi^B)}\right)}\\ &+\left(\exp\left(\frac{nq\left(1-\phi^B\right)}{(1-\alpha q)\phi^B(1-\phi)+1-\phi^B}\right)-1\right)\\ &\cdot\left(\frac{a}{\alpha q}+\frac{\phi^{B-1}(1-\phi)(1-\delta)}{\delta\left(1-\phi^B\right)}\right)+\frac{3}{2}-a.\end{aligned} \tag{17}$$

With an infinite energy buffer, nodes are more likely to sustain transmission. In this case, the following two corollaries provide analytical expressions for the network average AoI.

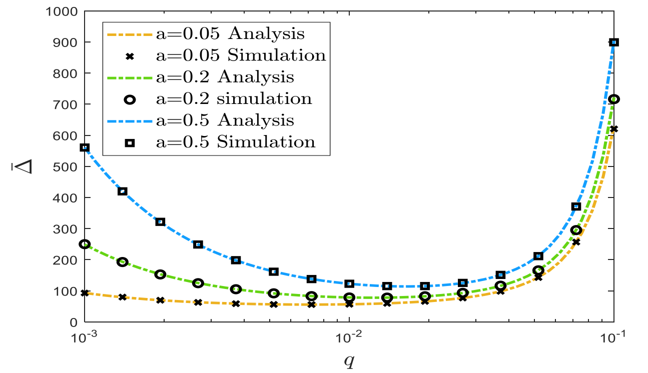


Fig. 4. The average AoI versus the packet generation rate q, $n = 40$.

Corollary 1. *When the energy buffer capacity $B \to \infty$, if $\delta \geq \frac{q}{q+a(1-q)+1-p}$, the network average AoI converges to*

$$\bar{\Delta}_{B\to\infty} = \frac{1+a-p}{pq} + \frac{3}{2} - \frac{3a}{2}, \tag{18}$$

in this case, the transmission success probability is given by

$$p = \exp\left(-nq\right). \tag{19}$$

Otherwise, if $\delta < \frac{q}{q+a(1-q)+1-p}$, the network average AoI converges to

$$\begin{aligned}\bar{\Delta}_{B\to\infty} =& \frac{\frac{(2-\alpha q)a^2}{(\alpha q)^2} + \frac{(\phi-1)(1-\delta)}{\phi}\left(\frac{2-a\delta}{\delta^2}+\frac{2a}{\alpha q\delta}\right)}{2\left(\frac{a}{\alpha q}+\frac{(\phi-1)(1-\delta)}{\delta\phi}\right)}\\ &+\left(\exp\left(\frac{nq}{(1-\alpha q)(\phi-1)+1}\right)-1\right)\\ &\cdot\left(\frac{a}{\alpha q}+\frac{(\phi-1)(1-\delta)}{\delta\phi}\right)+\frac{3}{2}-a,\end{aligned} \tag{20}$$

E. Average AoI Optimization

Building on the previous analysis, we aim to optimize the average AoI by determining the optimal packet generation rate. Specifically, we consider the scenario in which the energy buffer is infinite and formulate the optimization problem as

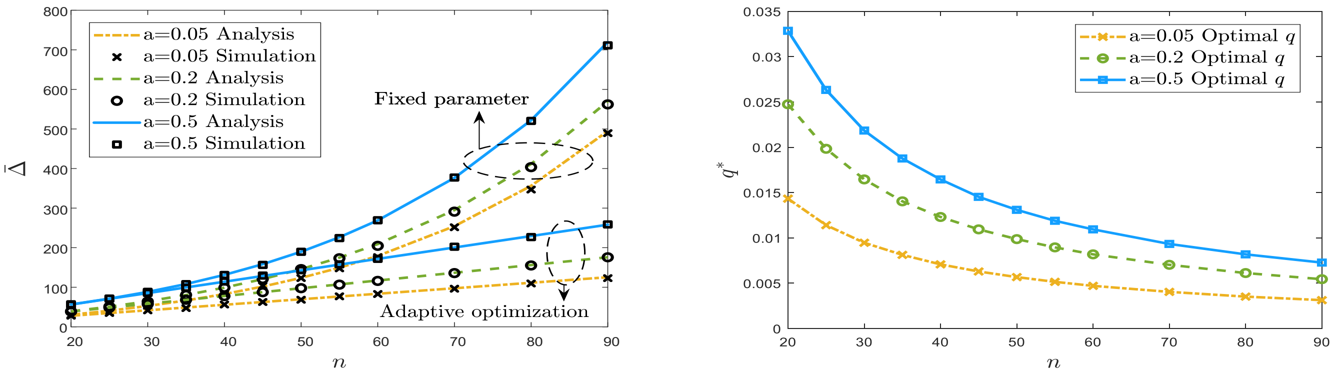
$$\begin{aligned}&\min_{\{q\}}\ \bar{\Delta}_{B\to\infty}\\ &\text{s.t.}\ \ q\in(0,1].\end{aligned} \tag{21}$$

According to the analysis in Corollary 1, under the assumption that the energy buffer capacity $B \to \infty$, the system operates in the energy-limited regime (**ELR**) if $\delta < \frac{q}{q+a(1-q)+1-p}$, and in the energy-sufficient regime (**ESR**) otherwise. We optimize the average AoI in each regime separately and then integrate the results for a comprehensive solution. Thus, the optimization problem is formulated as

$$\min_{\{q\}}\ \{\bar{\Delta}^{\mathbf{ELR}}, \bar{\Delta}^{\mathbf{ESR}}\}. \tag{22}$$

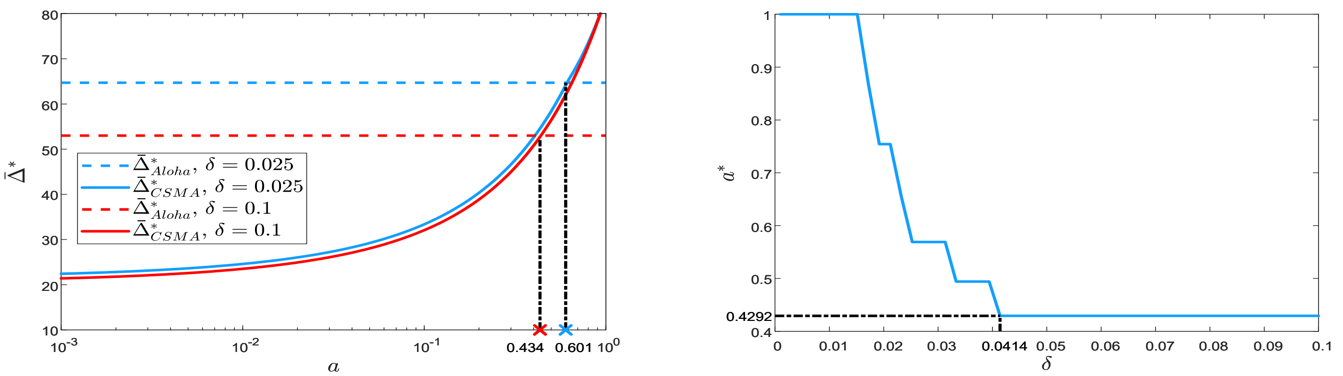
Theorem 1. *In the* **ESR**, *the corresponding optimal average AoI is given by*

$$\bar{\Delta}^{*,\mathbf{ESR}} = \frac{n}{-\mathbb{W}_0\left(-\frac{1}{e(1+a)}\right)} + \frac{3}{2} - \frac{3a}{2}, \tag{23}$$



(a) (b)

Fig. 5. (a) The average AoI with a fixed packet generation rate $q = 0.03$ compared to adaptive optimization that dynamically adjusts q. (b) The corresponding optimal packet generation rate versus the network size n.



(a) (b)

Fig. 6. (a) Optimal average AoI versus the mini-slot length a, $q = q^*$, $n = 20$, $\delta = 0.025$ or 0.1. (b) Mini-slot threshold of optimal average AoI a^* versus the energy arrival rate δ, $q = q^*$, $n = 20$.

which is achieved when the packet generation rate q is given by

$$q^{*,\mathbf{ESR}} = \frac{1}{n}\left[1 + \mathbb{W}_0\left(-\frac{1}{e(1+a)}\right)\right], \quad (24)$$

where $\mathbb{W}_0(\cdot)$ is the principal branch of the Lambert function.

Proof: According to (18) and (19), we have

$$\frac{d\bar{\Delta}}{dq} = \frac{\partial\bar{\Delta}}{\partial q} + \frac{\partial\bar{\Delta}}{\partial p}\frac{dp}{dq} = \frac{p - (1+a)(1-nq)}{pq^2}, \quad (25)$$

where $\frac{\partial\bar{\Delta}}{\partial q} = -\frac{1+a-p}{pq^2}$, $\frac{\partial\bar{\Delta}}{\partial p} = -\frac{1+a}{p^2 q}$ and $\frac{dp}{dq} = -np$. Let $g(q) = p - (1+a)(1-nq)$. It can be shown that $g(q)$ is strictly monotonically increasing with q, with $\lim_{q\to 0^+} g(q) = -a < 0$ and $g(1) > 0$. Therefore, there exists a unique value of q in the interval $(0, 1)$ at which $g(q) = 0$, indicating a sign change in $\frac{d\bar{\Delta}}{dq}$ from negative to positive. This implies that the optimal value of q satisfies $\frac{d\bar{\Delta}}{dq} = 0$. Finally, (24) follows by solving the equation $\frac{d\bar{\Delta}}{dq} = 0$. ■

In the **ELR**, the optimal average AoI, $\bar{\Delta}^{*,\mathbf{ELR}}$ and the optimal packet generation rate, $q^{*,\mathbf{ELR}}$, can be determined by using an exhaustive search algorithm, which is omitted here. Consequently, the optimal average AoI and the corresponding optimal packet generation rate for the CSMA network can be determined based on (22).

IV. Numerical Results

In this section, we validate the theoretical analysis through simulations. The simulation settings remain consistent with the system model, and each simulation runs for 10^6 time slots. Unless stated otherwise, we use the following parameter values: $\delta = 0.1$, $B = 200$, $a \in \{0.05, 0.2, 0.5\}$.

A. Analysis of Average AoI in CSMA Network

Fig. 4 illustrates the average AoI as a function of the packet generation rate under different mini-slot lengths. The close alignment between simulation and analysis validates the theoretical model. The average AoI first decreases and then increases with the generation rate, indicating a trade-off between update frequency and channel contention. A shorter mini-slot length a leads to a lower average AoI by allowing for more frequent channel detection and faster transmission decisions.

Fig. 5(a) compares the average AoI gain between a fixed packet generation rate and the proposed adaptive optimization under different mini-slot lengths. Compared with the network average AoI before optimization, the optimal average AoI increases linearly with network size n, while the growth rate slows as the mini-slot length a decreases. This indicates that enhanced carrier sensing capabilities, combined with adaptively optimized packet generation rates, can significantly reduce the network AoI. Fig. 5(b) presents the corresponding optimal packet generation rates q^*, showing that the optimal packet generation rate q^* decreases as the network size increases, effectively reducing collisions.

B. CSMA versus Aloha: Average AoI Comparison

So far, we have optimized the average AoI in EH-CSMA networks. Another widely used random access protocol, Aloha, can be similarly evaluated in terms of its average AoI. In prior work [9], we derived and optimized the average AoI for EH-Aloha networks. For consistency, we adopt the same assumptions as in the CSMA analysis: energy is harvested once per time slot with an arrival rate δ, and packets are managed by the generate-at-will strategy with update rate q. The closed-form expression for the optimal average AoI of slotted Aloha under the infinite energy buffer regime has already been derived in Theorem 3 of [9] and is omitted here for brevity. We compare the optimal performance of EH-CSMA and EH-Aloha networks under this infinite energy buffer regime. To standardize the notation, let $\bar{\Delta}^*_{\mathrm{CSMA}}$ and $\bar{\Delta}^*_{\mathrm{Aloha}}$ represent the minimum average AoI for the two networks, respectively.

Fig. 6(a) shows the optimal average AoI versus the mini-slot length a. As shown, $\bar{\Delta}^*_{\mathrm{Aloha}}$ remains constant due to Aloha does not employ carrier sensing and thus is independent of a. In contrast, $\bar{\Delta}^*_{\mathrm{CSMA}}$ benefits from smaller a, owing to frequent channel sensing that effectively reduces collisions, thereby improving the information freshness. However, as a increases, $\bar{\Delta}^*_{\mathrm{CSMA}}$ grows and eventually exceeds that of Aloha. This is attributed to the increased overhead of carrier sensing, which consumes channel resources without contributing to actual transmission. Furthermore, a higher energy arrival rate leads to better AoI performance in both protocols, as sufficient energy supply maintains timely transmission. These findings suggest that while CSMA offers advantages with a smaller mini-slot length a by leveraging channel awareness, Aloha becomes preferable under delay-tolerant scenarios due to its simpler access mechanism.

Fig. 6(b) illustrates how the critical mini-slot length a^* varies with the energy arrival rate δ. When $\delta < 0.0414$, nodes often become inactive due to limited energy, reducing the number of active nodes and the collisions. Even with longer carrier sensing caused by propagation delays, CSMA remains effective in avoiding collisions. Since AoI performance is mainly limited by energy rather than collision, CSMA shows higher tolerance to delay, resulting in a larger a^*. As $\delta \geq 0.0414$, the energy buffer is usually non-empty, enabling continuous transmissions. Thus, excessively long sensing durations may delay transmissions. Consequently, a^* drops significantly to 0.4292, indicating that CSMA only outperforms Aloha when the mini-slot length a is smaller than a^*. These results offer theoretical insights for protocol selection in EH networks. It is worth noting that, since carrier sensing also consumes a small amount of energy, the actual range in which CSMA outperforms Aloha may be narrower than the theoretical prediction.

V. Conclusion

We derive a closed-form expression for the average AoI in EH-CSMA networks with arbitrary energy buffer capacity. Based on this result, we obtain the optimal average AoI and the corresponding optimal packet generation rate for CSMA networks with an infinite energy capacity. Furthermore, we compare the optimal average AoI of EH-CSMA and EH-Aloha networks, revealing that CSMA outperforms Aloha when the mini-slot length is small, due to its effective collision avoidance enabled by frequent channel sensing. However, as the mini-slot length increases and indicates higher propagation delay and sensing overhead, Aloha becomes more favorable due to its contention-free access mechanism.

References

[1] S. Kaul, R. Yates, and M. Gruteser, “Real-time status: How often should one update?,” in *Proc. IEEE INFOCOM*, pp. 2731–2735, 2012.
[2] A. Kosta, N. Pappas, and V. Angelakis, *Age of Information: A New Concept, Metric, and Tool.* 2017.
[3] L. Li, Y. Dong, C. Pan, and P. Fan, “Age of Information of CSMA/CA Based Wireless Networks,” in *Proc. IEEE IWCMC*, pp. 530–535, 2022.
[4] A. Maatouk, M. Assaad, and A. Ephremides, “On the Age of Information in a CSMA Environment,” *IEEE ACM Trans. Netw.*, vol. 28, pp. 818–831, Feb. 2020.
[5] M. Moltafet, M. Leinonen, and M. Codreanu, “Worst Case Age of Information in Wireless Sensor Networks: A Multi-Access Channel,” *IEEE Wireless Commun. Lett.*, vol. 9, pp. 321–325, Nov. 2020.
[6] F. Zhao, X. Sun, W. Zhan, X. Wang, J. Gong, and X. Chen, “Age-Energy Tradeoff in Random-Access Poisson Networks,” *IEEE Trans. Green Commun. Netw.*, vol. 6, no. 4, pp. 2055–2072, 2022.
[7] A. Yuan, F. Zhao, and X. Sun, “Energy-Aware Random Access Networks: Connection-Based Versus Packet-Based,” *IEEE Commun. Lett.*, vol. 28, no. 9, pp. 2216–2220, 2024.
[8] K.-H. Ngo, G. Durisi, A. G. Amat, A. Munari, and F. Lázaro, “Age of Information in Slotted ALOHA With Energy Harvesting,” in *Proc. IEEE GLOBECOM*, pp. 3579–3584, 2023.
[9] S. Xiao, X. Sun, W. Zhan, and X. Wang, “Information Freshness in Random Access Networks with Energy Harvesting,” in *Proc. IEEE ITW*, pp. 127–132, 2024.
[10] F. Zhao, N. Pappas, M. Zhang, and H. H. Yang, “Age of Information in Random Access Networks with Energy Harvesting,” *Available as ArXiv: 2412.01192*, Dec. 2024.
[11] G. Yang, G.-Y. Lin, and H.-Y. Wei, “Markov chain performance model for ieee 802.11 devices with energy harvesting source,” in *Proc. IEEE GLOBECOM*, pp. 5212–5217, 2012.
[12] L. Dai, “Toward a Coherent Theory of CSMA and Aloha,” *IEEE Trans. Wireless Commun.*, vol. 12, pp. 3428–3444, Jun. 2013.
[13] X. Sun and L. Dai, “To Sense or Not To Sense: A Comparative Study of CSMA With Aloha,” *IEEE Trans. Commun.*, vol. 67, pp. 7587–7603, Jul. 2019.
[14] D. Wu, W. Zhan, X. Sun, and J. Qiu, “Peak Age of Information Optimization: CSMA Versus Aloha,” in *IoT as a Service*, pp. 436–450, 2025.
[15] X. Lu, P. Wang, D. Niyato, D. I. Kim, and Z. Han, “Wireless Networks With RF Energy Harvesting: A Contemporary Survey,” *IEEE Commun. Surveys Tuts.*, vol. 17, no. 2, pp. 757–789, Nov. 2015.
[16] R. J. M. Vullers, R. V. Schaijk, H. J. Visser, J. Penders, and C. V. Hoof, “Energy Harvesting for Autonomous Wireless Sensor Networks,” *IEEE Solid-State Circuits Mag.*, vol. 2, no. 2, pp. 29–38, Feb. 2010.
[17] J. Polastre, J. Hill, and D. Culler, “Versatile low power media access for sensor networks,” in *Proc. 2nd Int. Conf. Embedded Netw. Sensor Syst.*, vol. 1, pp. 95–107, 2004.
[18] Texas Instruments, “CC2420: 2.4 GHz IEEE 802.15.4 / ZigBee-Ready RF Transceiver.” Data Sheet, 2007. https://www.ti.com/lit/gpn/cc2420.